Samuele Sparno

Caserta, Italy | 15/01/2003 | +39 3407277081 | sparno.samuele70@gmail.com | Github | Linkedin

EDUCATION

Master's Degree in Cybersecurity – University of Salerno — Salerno, Italy
2025 – Present

Bachelor's Degree in Computer Science – University of Salerno — Salerno, Italy
Final Grade: 98/110 — 2021 – 2025

- **Key Courses:** Computer Networks, Cybersecurity, Web Development, Database Management, Object-Oriented Programming, Foundations of Artificial Intelligence

Elements of AI - University of Helsinki & MinnaLearn — Online
- Fundamentals of Machine Learning and Neural Networks, Future Implications of Artificial Intelligence — 2021

High School Diploma in Computer Science ITI-LS "F.Giordani" — Caserta, Italy
Final Grade: 100/100 — 2016 – 2021

WORK EXPERIENCE

Internship - Network & Infrastructure Monitoring
System Management S.p.A. — 12/2024 – 03/2025
- Developed the Bachelor's thesis project: *"Design and Implementation of a Network with a Monitoring System"*

Private Tutor — 2022 – 2024
- Tutoring in basic programming (C language) and database management lessons

Freelance VideoEditor — 2020 – 2024
- Created commissioned videos, promotional content, and memorable videos for commercial activities

PROJECTS

Secure IoT Camera Pipeline for Forensic Integrity Verification — Github Repo
- Developed as part of a 2-person team an IoT camera pipeline for motion-triggered capture, encrypted transmission, integrity verification, and blockchain notarization.
- Focused on the security/backend side: encrypted data processing, AES encryption, SHA-256 integrity proofs, and IPFS + Ethereum hash notarization.
- Implemented digital-signature device authentication, device identity verification, duplicate-record checks, and nonce-based replay protection.
- Used blockchain to store photo hashes for forensic integrity checks, tamper evidence, and traceability.
- Technologies Used: Go, C/C++, Solidity, Python, MQTT, AES, SHA-256, IPFS, Hyperledger FireFly, Ethereum, REST APIs, Git.

Implementation of a Network with a Monitoring System — Github Repo
- Developed during an internship at *System Management S.p.A.*: designed and implemented a network infrastructure with a monitoring system based on Nagios Core, using real physical devices.
- Implemented host/service monitoring (HTTP, FTP), system resource checks with NSClient++, and automated email alerting.
- Simulated real-world fault scenarios to validate monitoring coverage and proactive alert management.
- Technologies Used: GNS3, FortiGate, VMware, Cisco Routers & Switches, Nagios Core, NSClient++, Postfix

TalkAID — Github Repo
- Developed in a structured team environment with role-based task division and coordination with project managers, for a web platform supporting remote speech therapy sessions.
- Contributed to user profile management, patient exercises, session booking, and progress tracking features.
- Integrated an AI module for exercise recommendations and Azure AI for voice evaluation.
- Technologies Used: HTML, CSS, JavaScript, Java, SQL, Git

TECHNICAL SKILLS

Programming Languages: Python, Java, C/C++, JavaScript, SQL, Go, Solidity
Cybersecurity: Security Monitoring, Log Analysis, Network Security, Digital Forensics, Cryptography, Integrity Verification, Replay Protection
Infrastructure & Tools: Nagios Core, NSClient++, Elasticsearch, Grafana, Postfix, GNS3, FortiGate, VMware, Git, Docker
Operating Systems: Linux, Windows

SKILLS

Languages: Italian (Native), English (B2 – Upper-Intermediate)
Driver's License: Category B (Car)

I hereby authorize the processing of my personal data included in my CV in accordance with Article 13 of Legislative Decree No. 196 of June 30, 2003 – 'Personal Data Protection Code' – and Article 13 of EU Regulation 679/2016 – 'General Data Protection Regulation (GDPR)'

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